# OpenVis: Real-Time Observability Tool for OpenFlow SDN Networks

Rakhat Yskak

Department of Computer Science
School of Engineering and Digital Sciences
Nazarbayev University

November 2025

NAZARBAYEV UNIVERSITY

# The Observability Challenge in SDN

- SDN centralizes control, but creates a **visibility gap**
- Operators need to see what controllers are doing
- Topology changes happen dynamically
- Multiple controllers add complexity

# Introducing OpenVis

## What is OpenVis?

Web-based observability tool for OpenFlow SDN networks with real-time visualization

## Key Features

- Live topology visualization
- Color-coded link utilization
- Multi-controller monitoring
- Flow statistics tracking
- Network events timeline




Figure: Interactive force-directed graph

NAZARBAYEV UNIVERSITY

# System Architecture

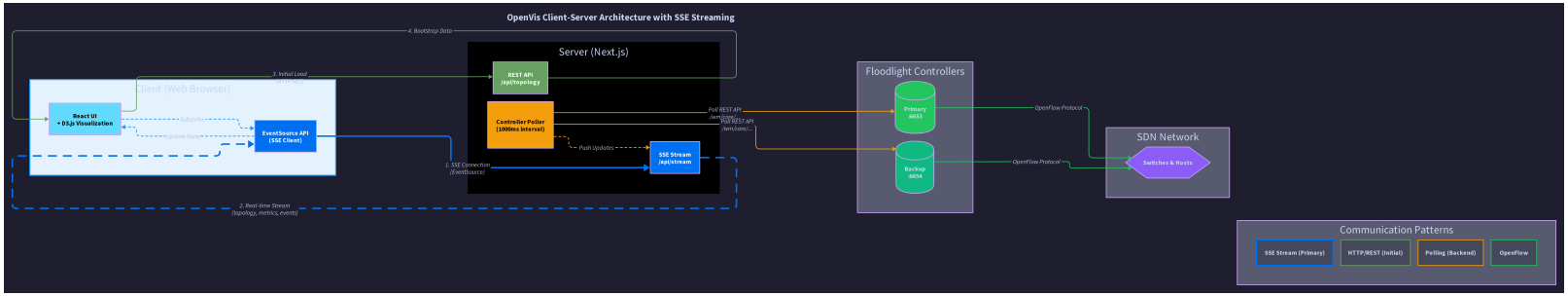



Figure: Client-server architecture with SSE streaming

## Technology Stack

- **Frontend:** React + TypeScript + D3.js force-directed graphs
- **Backend:** Next.js with Server-Sent Events (SSE)
- **Controllers:** Floodlight via standard OpenFlow REST API
- **Key Innovation:** Real-time updates without polling from client

# Core Capabilities

## 1. Real-Time Topology Visualization

- D3.js force-directed graph with automatic layout
- Interactive: drag nodes, zoom, pan
- Color coding: green (controllers), blue (switches), purple (hosts)

## 2. Traffic Monitoring

- Links colored by utilization: green (low) $\rightarrow$ yellow (medium) $\rightarrow$ red (high)
- Bottlenecks become immediately visible
- Port statistics and bandwidth metrics

## 3. Multi-Controller Support

- Monitor multiple controllers simultaneously
- Health status indicators with auto-reconnect
- Failover detection

# Technical Implementation

## Server-Sent Events (SSE) Streaming

- Backend polls Floodlight REST API (configurable interval)
- Server pushes updates to clients via SSE
- Client receives real-time topology changes without refresh
- Automatic error handling and retry logic

## Flow Statistics & Events

- Track flow counts, packet/byte counters per switch
- TCAM capacity monitoring with warnings
- Timeline of network events: links up/down, switch/host changes

# Testing and Validation

## Test Environment

- **Emulation:** Mininet network emulator
- **Controllers:** Floodlight in Docker containers
- **Traffic Generation:** iperf for bandwidth testing

## 8 Test Scenarios

- Simple topology (3 switches)
- Linear topology (scalability)
- Tree topology (hierarchical)
- Traffic load testing
- Multi-controller failover
- Link failure recovery
- Congestion patterns (elephant flows, hotspots)
- Flow table monitoring

All tests passed: topology changes, controller states, and traffic patterns visualized correctly

# Conclusion

A web-based real-time observability tool that helps network operators understand SDN behavior through:

- Interactive topology visualization (D3.js)
- Live traffic monitoring (SSE streaming)
- Multi-controller health tracking
- Flow statistics and event timeline

# Conclusion

A web-based real-time observability tool that helps network operators understand SDN behavior through:

- Interactive topology visualization (D3.js)
- Live traffic monitoring (SSE streaming)
- Multi-controller health tracking
- Flow statistics and event timeline

## Future Enhancements

- Historical data storage and trend analysis
- Support for ONOS and OpenDaylight controllers
- Alert notifications for critical events
- Machine learning for anomaly detection

# Thank You

**Rakhat Yskak**
Department of Computer Science
Nazarbayev University

**GitHub Repository:**
`https://github.com/4ry1337/sdn/tree/main/Project`

## Questions?